

Problem 1:-

$$V = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

① Zero vector (additive identity)

$$(x, y, z) + \vec{0} = (x+0, y+0, z+0) = (x, y, z)$$

→ Additive inverse

$$\vec{u}(-\vec{u}) = \vec{0}$$

$$(x, y, z) + (-x, -y, -z) = (x-x, y-y, z-z) = (0, 0, 0)$$

So with these operation it is
vector space

(b)

(b) colors intensity can not go in negative so color range is $c=1$

$$(r_1, g_1, b_1) \oplus (r_2, g_2, b_2) = \min(r_1 + r_2, 1) + \min(g_1 + g_2, 1) + \min(b_1 + b_2, 1)$$

$$V = \{v \mid v = (c \cdot a, 1, c)\}$$

It's additive inverse can't exist because $-v = (-c \cdot a, -1, c)$ don't exist in V

axiom 5 failed $\vec{v} + (-v) = \vec{0}$

If we take $k=2$

$$v = (c \cdot a, 1, c)$$

$$kv = (c \cdot 4, 2, c) \notin V$$

axiom 6 failed

Problem 22.

$$(b) W = \{ (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid x_1 + x_2 + x_3 + x_4 = 1 \}$$

(1) does H contains zero vector -

by setting all points to zero $0=1$

so H is not satisfying

so H is not subspace of $\mathbb{R}^4$

For suppose it's additive inverse

$$-x_1 - x_2 - x_3 - x_4 = -1 \sim \text{so out of bound}$$



$$(c) W_3 = \{ (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid x_1 = 2x_3 \text{ and } x_2 = x_4 \}$$

(1) does it contains zero vector

$$\text{if } x_1 = 0, x_2 = 0 \\ \text{and } x_4 = 0, x_3 = 0$$

Yes satisfying

(2) does - if $x_3 = 1, x_4 = 1$ then

$$x_1 = 2, x_2 = 1$$

$$\text{so } (x_1, x_3) = (2, 1) \text{ and } x_3' = (2) \text{ then } x_1' = 4$$

$$(x_2, x_4) = (1, 1) \text{ and } x_4' = 2, x_2' = 2$$

$$\text{so } (x_1, x_3) + (x_1', x_3') = (x_1 + x_1', x_3 + x_3') \in W_3$$

$$(x_2, x_4) + (x_2', x_4') = (x_2 + x_2', x_4 + x_4') \in W_3$$

$$(2, 1) + (4, 2) = (6, 3), x_1 = 6, x_3 = 3$$

$$(1, 1) + (2, 2) = (3, 3), x_2 = 3, x_4 = 3$$

so satisfying closure under addition

$$(3) u = (x_1, x_3) = (2, 1)$$

$$v = (x_2, x_4) = (1, 1)$$

$$kv = (4, 2) \text{ so it is closure under scalar multiplication}$$

$$kv = (2, 2)$$

so it is subspace of $\mathbb{R}^4$

Problem 3g-

$$\text{Span}(A_1, A_2, A_3, A_4) = M_{22}$$

$$(1) \begin{bmatrix} a & b \\ c & d \end{bmatrix} = k_1 A_1 + k_2 A_2 + k_3 A_3 + k_4 A_4$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = k_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + k_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + k_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + k_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} k_1 & k_2 \\ k_3 & k_4 \end{bmatrix} \quad \text{so} \quad \begin{aligned} k_1 &= a \\ k_2 &= b \\ k_3 &= c \\ k_4 &= d \end{aligned}$$

∴ These are standard basis vectors of M_{22} and can generate any possible matrix

$$a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

(2) If $B = \begin{bmatrix} 3 & -2 \\ 5 & 4 \end{bmatrix}$ as we saw on part (1) we only need scalars to

form any matrix so

$$\begin{bmatrix} 3 & -2 \\ 5 & 4 \end{bmatrix} = 3 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + (-2) \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 5 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Problem 48-

$$1// \quad v_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$$

$$v_4 = \begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix}$$

1st 0 if any vector space is linearly independent
mean H has only trivial solution

$$V = \{v_1, v_2, v_3, v_4\}$$

$$k_1 v_1 + k_2 v_2 + k_3 v_3 + k_4 v_4 = \vec{0}$$

$$k_1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + k_2 \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} + k_3 \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} + k_4 \begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix} = \vec{0}$$

$$\begin{bmatrix} k_1 & 3k_2 & 2k_3 & 0k_4 \\ 2k_1 & k_2 & -k_3 & 5k_4 \\ -k_1 & 4k_2 & 5k_3 & 2k_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

write in augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 3 & 2 & 0 & 0 \\ 2 & 1 & -1 & 5 & 0 \\ -1 & 4 & 5 & 2 & 0 \end{array} \right]$$

$$-2R_1 + R_2 \rightarrow R_2$$

$$R_1 + R_3 \rightarrow R_3$$



$$\left[\begin{array}{cccc|c} 1 & 3 & 2 & 0 & 0 \\ 0 & -5 & -5 & 5 & 0 \\ 0 & 7 & 7 & 2 & 0 \end{array} \right]$$

$$-1x_0 \rightarrow R_2$$

$$\left[\begin{array}{cccc|c} 1 & 3 & 2 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 7 & 7 & 2 & 0 \end{array} \right]$$

$$-3x_0 + x_1 \rightarrow R_1$$

$$-7x_0 + x_3 \rightarrow R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 3 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 9 & 0 \end{array} \right]$$

$$\frac{1}{9}x_4 \rightarrow R_4$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 3 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

$$x_4 + x_3 \rightarrow R_3$$

$$-3x_4 +$$

$$-x_3 + x_2 \rightarrow R_2$$

$$-3x_3 + x_1 \rightarrow R_1$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 3 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

$$x_1 - x_3 = 0$$

$$x_2 + x_3 = 0$$

$$x_4 = 0$$

Where x_3 is

free variable so it is not linearly independent



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$$2, \quad k_2 = t$$

$$, \quad k_1 = +t$$

$$k_0 = -t$$

$$k_4 = 0$$

$$k_1 \vec{v}_1 + k_0 \vec{v}_0 + k_3 \vec{v}_3 + k_4 \vec{v}_4 = \vec{0}$$

$$t \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} - t \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$3, \quad W = \int_a^b \begin{vmatrix} \cancel{c} & \begin{matrix} v_1 \\ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{matrix} & \begin{matrix} v_0 \\ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \end{matrix} & \begin{matrix} v_3 \\ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \end{matrix} \end{vmatrix} dx$$

$$k_1 \vec{v}_1 + k_0 \vec{v}_0 + k_3 \vec{v}_3 = \vec{0}$$

$$\begin{bmatrix} 1 & 0 & a & c \\ 0 & 1 & b & c \\ 0 & 0 & c & 0 \end{bmatrix} \quad \begin{array}{l} k_1 + a k_3 = 0 \\ k_0 + b k_3 = 0 \\ k_3 c = 0 \end{array}$$

we will check by determinant if
 $\det(\text{matrix}) \neq 0$ so linearly independent

$$\begin{vmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & c \end{vmatrix} = +1(c-0) + 0() + a(c-0) \neq 0$$

$c \neq 0$ then it is linearly independent

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Problem 5

$$1/ \quad A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$$

c_1, c_2, c_3, c_4 are coordinates of vector A relative to B

$$c_1 T_1 + c_2 T_2 + c_3 T_3 + c_4 T_4 = A$$

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$+ c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$$

$$[A]_B = \begin{bmatrix} 3 \\ 1 \\ 2 \\ 4 \end{bmatrix}$$

$$c_1 = 3$$

$$c_2 = 1$$

$$c_3 = 2$$

$$c_4 = 4$$

$$c_1 + c_2 + c_3 + c_4 = 3$$

$$c_1 + c_2 + c_3 + c_4 = 1$$

$$c_1 + c_2 + c_3 + c_4 = 2$$

$$c_1 + c_2 + c_3 + c_4 = 4$$

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Problem 5 //

Q. $P: B \rightarrow B'$ from basis B to basis B'

$P: B \rightarrow B'$

Gauss Jordan elimination

$[B' | B] \rightarrow [I_n | P^{-1}]$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 1 \end{array} \right]$$

$-R_1 + R_2 \rightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 1 \end{array} \right]$$

$\frac{-1}{2} R_2 \rightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 1 \end{array} \right]$$

$-R_2 + R_1 \rightarrow R_1$

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Problem 5" 2, Continue

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/2 & +1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 1 \end{array} \right]$$

$-R_3 + R_4 \rightarrow R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/2 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 & -1 & 1 \end{array} \right]$$

$-\frac{1}{2}R_4 \rightarrow R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/2 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1/2 & -1/2 \end{array} \right]$$

$-R_4 + R_3 \rightarrow R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/2 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & +1/2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1/2 & -1/2 \end{array} \right]$$

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Problem 54

2 Continue

$$P^{-1} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & -1/2 \end{bmatrix}$$

F

3 $[A]_{B'} = P^{-1} [A]_B$

$$[A]_{B'} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 2 \\ 4 \end{bmatrix}$$

$$[A]_{B'} = \begin{bmatrix} 3/2 + 1/2 + 0 + 0 \\ 3/2 - 1/2 + 0 + 0 \\ 0 + 0 + 1 + 2 \\ 0 + 0 + 1 - 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$

Part 9

$$x_1 + 2x_2 - x_3 + 3x_4 - 2x_5 = 0$$

$$2x_1 + 4x_2 - 2x_3 + 7x_4 - 3x_5 = 0$$

$$3x_1 + 6x_2 - 3x_3 + 10x_4 - 5x_5 = 0$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 3 & -2 & 0 \\ 2 & 4 & -2 & 7 & -3 & 0 \\ 3 & 6 & -3 & 10 & -5 & 0 \end{array} \right]$$

$$-2r_1 + r_2 \rightarrow R_2$$

$$-3r_1 + r_3 \rightarrow R_3$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 3 & -2 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array} \right]$$

$$-r_2 + r_3 \rightarrow R_3$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 3 & -2 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\# \quad c_1 + 2c_2 - c_3 + 3c_4 - 2c_5 = 0$$

$$c_4 + c_5 = 0$$

c_2, c_3, c_4, c_5 are free variables

$$\left[\begin{array}{l} c_1 = -2c_2 + c_3 - 3c_4 + 2c_5 \\ c_4 = -c_5 \end{array} \right]$$

$$\therefore c_2 = 1, c_3 = 5, c_4 = 4, c_5 = -4$$

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} -2t + s - 3u - 2v \\ t \\ s \\ u \\ -u \end{bmatrix}$$

Separate these three parameters

$$S = \{ (-2t, t, 0, 0, 0) + (s, 0, s, 0, 0) + (-3u - 2v, 0, 0, u, -u) \}$$

$$S = \{ (-2, 1, 0, 0, 0), (1, 0, 1, 0, 0), (-3, -2, 0, 1, -1) \}$$

2" $\dim(S) = 3$

3" ~~$x_1 + 2x_2 - x_3 + 3x_4 - 2x_5 = 0$~~
 ~~$2x_1 + 4x_2 - 2x_3 + 7x_4 - 3x_5 = 0$~~
 ~~$3x_1 + 6x_2 - 3x_3 + 10x_4 - 5x_5 = 0$~~

3" $x_1 + x_2 + x_3 = 0$



$$x_1 = -2t + s - 5u$$

$$x_2 = t, x_3 = s$$

$$-2t + s - 5u + t + s = 0$$

$$-t + 2s - 5u = 0$$

$$t = 2s - 5u$$



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Put t in equations

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} -2t + s - 5u \\ 6 \\ s \\ 4 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} -4s + 10u + s - 5u \\ 2s - 5u \\ s \\ u \\ -4 \end{bmatrix} = \begin{bmatrix} 4s \\ 2s - 5u \\ s \\ u \\ -4 \end{bmatrix}$$

now separate these two parameters

$$S = \int \begin{pmatrix} 4s \\ 2s - 5u \\ s \\ u \\ -4 \end{pmatrix} + \begin{pmatrix} s \\ -5u \end{pmatrix}$$

$$S = \mathcal{L} \left((u) - 5u, 0, u, -u \right) + (s, 2s, s, 0, 0)$$

$$S = \mathcal{L} \left((1, -5, 0, 1, -1), (1, 2, 1, 0, 0) \right)$$

$$\boxed{\dim(S) = 2}$$



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1) W is set of all 3×3 symmetric matrices

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 5 & 4 \\ 1 & 4 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \quad B^T = B$$

$$A^T = A$$

We know by adding two symmetric matrices, it gives symmetric matrix

$$\rightarrow A+B = R$$

$$R = \begin{bmatrix} 3 & 5 & 4 \\ 5 & 9 & 10 \\ 4 & 10 & 15 \end{bmatrix} \quad R^T = R$$

So H is closure under addition

It also H is closure under

$$H = 2$$

scalar multiplication

$$H \begin{bmatrix} 2 & 3 & 1 \\ 3 & 5 & 4 \\ 1 & 4 & 6 \end{bmatrix} = \begin{bmatrix} 4 & 6 & 2 \\ 6 & 10 & 8 \\ 2 & 8 & 12 \end{bmatrix} \quad \text{this is symmetric}$$

dim If W is subspace of $M_{3,3}$ then
its $\dim(W) = 9$



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2) $p(1)=0, p(-1)=0$

$$P_4 = a + a_1x + a_2x^2 + a_3x^3 + a_4x^4$$

$$P_4(1) = a + a_1 + a_2 + a_3 + a_4 = 0 \quad \text{--- (1)}$$

$$P_4(-1) = a - a_1 + a_2 - a_3 + a_4 = 0 \quad \text{--- (2)}$$

Add these equation

$$2a + 2a_2 + 2a_4 = 0$$

$$\boxed{a + a_2 + a_4 = 0} \quad \therefore a = -a_2 - a_4$$

Now subtract both equation

$$a + a_1 + a_2 + a_3 + a_4 = 0$$

$$-a + a_1 - a_2 + a_3 - a_4 = 0$$

$$a_1 + a_3 = 0 \rightarrow a_1 = -a_3$$

$$P_4 = a + a_1x + a_2x^2 + a_3x^3 + a_4x^4$$

$$P_4 = (-a_2 - a_4) + (-a_3)x + a_2x^2 + a_3x^3 + a_4x^4$$

$$P_4(1) = \cancel{-a_2} - \cancel{a_4} - \cancel{a_3} + \cancel{a_2} + \cancel{a_3} + \cancel{a_4} = 0$$

$$P_4(-1) = \cancel{(-a_2 - a_4)} + \cancel{a_3} + \cancel{a_2} - \cancel{a_3} + \cancel{a_4} = 0$$

so it is satisfying both property
so it is subspace of P_4



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Q6" continue

$\dim(P_4) = \underline{5}$ $\dim(P_4) \rightarrow$ We have 3 dependent
coeff so $\rightarrow \underline{3}$

$$P_4 = (-q_0 - q_4) + (-q_3)x + q_0x^2 + q_3x^3 + q_4x^4$$

↓

$$P_4 = -q_0 + q_0x^2 - q_3x + q_3x^3 + q_4x^4 - q_4$$

$$= q_0(-1 + x^2) - q_3(-x + x^3) + q_4(x^4 - 1)$$

so bases are $\{(-1 + x^2), (-x + x^3), (x^4 - 1)\}$

$$\frac{-13+4}{2} = -7.5+4 = -3.5 = -\frac{7}{2}$$

$$\frac{-6.5+4}{-2.5} = -\frac{2.5}{-2.5} = 1$$

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Q.7

$$P^{-1}: B \rightarrow B' \quad [B'|B] \rightarrow [I_n|P^{-1}]$$

$$\left[\begin{array}{cc|cc} 1 & -1 & 2 & 4 \\ 3 & -1 & 2 & -1 \end{array} \right]$$

$$-3R_1 + R_2 \rightarrow R_2$$

$$\left[\begin{array}{cc|cc} 1 & -1 & 2 & 4 \\ 0 & 2 & -4 & -13 \end{array} \right]$$

$$R_2 \rightarrow R_2/2$$

$$\left[\begin{array}{cc|cc} 1 & -1 & 2 & 4 \\ 0 & 1 & -2 & -13/2 \end{array} \right]$$

$$R_1 + R_2 \rightarrow R_1$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 0 & -5/2 \\ 0 & 1 & -2 & -13/2 \end{array} \right]$$

$$P^{-1} = \begin{bmatrix} 0 & -2/2 \\ -2 & -13/2 \end{bmatrix} \rightarrow \text{Transition matrix}$$

$$[W]_{B'} = P^{-1} [W]_B = \begin{bmatrix} 0 & -2/2 \\ -2 & -13/2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

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$$\begin{aligned} &= 0 + 7/2 = \overset{S}{\cancel{7/2}} \quad F \\ &= +2 - 13 = \begin{bmatrix} \cancel{7/2} \\ -11 \end{bmatrix} \quad F \end{aligned}$$

$$\cancel{[W]_B} = \cancel{\begin{bmatrix} 7/2 \\ -11 \end{bmatrix}}$$

$$= \cancel{F} \begin{bmatrix} 0 & -5 \\ +2 & -13 \end{bmatrix} = \begin{bmatrix} -5 \\ -11 \end{bmatrix}$$

Q.8 $A = \begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & -1 & 7 & 7 \\ -1 & 3 & -4 & 2 & -5 & -4 \end{bmatrix}$

$R = \text{RREF}(A) = \begin{bmatrix} 1 & -3 & 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

(1) Basis for row space, $\text{Row}(A)$

The non zero rows of RREF form a basis for row space



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Basis for $\text{row}(A)$: $\left\{ \begin{bmatrix} 1 & -3 & 0 & 0 & 3 \end{bmatrix}, \right.$

$\begin{bmatrix} 0 & 0 & 1 & 0 & -2 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & -2 & 0 \end{bmatrix}$
 $\left. , \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \right\}$

So $\dim(\text{row}(A)) = 4$

(2) Basis for column space, $\text{col}(A)$

Basis for $\text{col}(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 9 \\ 9 \\ -4 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ -1 \\ 2 \end{bmatrix} \right.$
 $\left. , \begin{bmatrix} 4 \\ 2 \\ 7 \\ -4 \end{bmatrix} \right\}$

$\dim(\text{col}(A)) = 4$

(3) Basis for Null space, $\text{null}(A)$

$\begin{cases} x_1 - 3x_2 + 3x_5 = 0 \\ x_3 - 2x_5 = 0 \\ x_4 + x_5 = 0 \\ x_6 = 0 \end{cases}$

$x_1 = 3x_2 - 3x_5$

$x_3 = 2x_5$

$x_4 = -x_5$

$x_6 = 0$



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$$x = x_0 \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} -3 \\ 0 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} \rightarrow \text{These are} \\ \text{basis of} \\ \text{null}(A)$$

$$\dim(\text{null}(A)) = 2$$

4.

$$y'' \quad b = \begin{bmatrix} 1 \\ -9 \\ -5 \\ -4 \end{bmatrix}$$

$$\left[\begin{array}{cccc|c} 1 & 4 & -2 & 4 & 1 \\ 2 & 9 & -1 & 2 & -9 \\ 2 & 9 & -1 & 7 & -5 \\ -1 & -4 & 2 & -4 & 4 \end{array} \right]$$

$R_4 + R_1$

$$\left[\begin{array}{cccc|c} 1 & 4 & -2 & 4 & 1 \\ 2 & 9 & -1 & 2 & -9 \\ 2 & 9 & -1 & 7 & -5 \\ 0 & 0 & 0 & 0 & 5 \end{array} \right]$$

o/s

so system is inconsistent. and b is not
column space of this matrix



Problem-9

1/ For largest value of $\text{rank}(A)$

$$\text{rank}(A) \leq \min(5, 7)$$

$$\text{rank}(A) \leq 5$$

so largest value of $\text{rank}(A) = 5$

$$\text{rank}(A) + \text{nullity}(A) = 7$$

Now smallest of nullity we must take largest value of rank

$$\text{so smallest nullity} = 7 - 5 = 2$$

(a)

(a) free variable = nullity of A

$$\text{free variable} = n - \text{rank}(A)$$

$$= 7 - 4$$

$$\boxed{= 3}$$

(b) The solution is not unique because $\text{nullity}(A) \neq 0$

$$\text{3- (a) } \text{rank}(A) = n - \text{nullity}(A)$$

$$\text{rank}(A) = 7 - 2 = 5$$

(b) $\text{rank}(A)$ is 5 and $\dim(\mathbb{R}^5) = 5$ so

the column space is a subspace of $\mathbb{R}^5$

and it spans $\mathbb{R}^5$

